

1. 求 $w = z^2$ 在 $z = i$ 处的伸缩率和旋转角. 问: $w = z^2$ 将经过点 $z = i$ 并平行于实轴正向的曲线的切线方向映射成 w 平面上哪一个方向? 并作图.

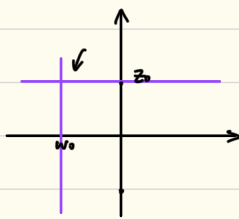
解: $(z^2)'|_{z=i} = 2i$

$\therefore$ 伸缩率 $|2i| = 2$

转动角 $\text{Arg } 2i = \frac{\pi}{2}$

$z_0 = i \quad w_0 = -1$

$\therefore$ 平行于虚轴方向 $z = -1 + yi, y \in \mathbb{R}$



$|f'(z_0)|$
 $\text{arg } f'(z_0)$

2. 映射 $w = z^2$ 把上半圆域 $|z| < R, \text{Im}(z) > 0$ 映射成什么图形

解: 令 $z = re^{i\theta}, w = \rho e^{i\varphi}$,

$\therefore \rho = r^2, \varphi = 2\theta$

$\therefore 0 < \theta < \pi$

$\therefore 0 < \varphi < 2\pi$

$\therefore$ 映射成圆心在原点, 半径为 R^2 , 且沿由 0 到 R^2 的半径有割痕的圆域

3. 下列区域在指定映射下映射成什么? 设出复数

(1) $\text{Im}(z) > 0, w = (1+i)z$

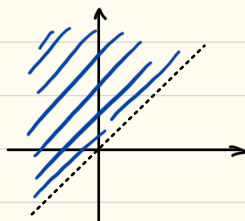
解: 设 $z = x + iy, w = u + iv$

$\therefore u + iv = (1+i)(x+iy) = x-y + (x+y)i$

$\therefore u = x-y, v = x+y$

$\therefore y > 0$

$\therefore v > u, \text{Im}(w) > \text{Re}(w)$



(2) $0 < \text{Im}(z) < \frac{1}{2}, w = \frac{1}{z}$

解: 反演变换具有保圆性

边界 $C_1: y = 0, C_2: y = \frac{1}{2}$.

将 $0, 1, \infty$ 映射成 $\infty, 1, 0$

设 $z = x + iy, C_1: w = \frac{1}{x} (\text{Im}(z) = 0)$ 即 $\text{Im}(w) = 0$.

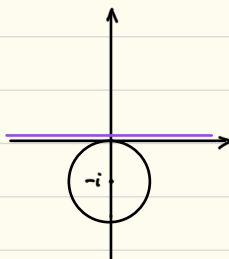
$C_2: w = \frac{x - \frac{1}{2}i}{x^2 + \frac{1}{4}}, u = \frac{x}{x^2 + \frac{1}{4}}, v = \frac{-\frac{1}{2}}{x^2 + \frac{1}{4}}, u^2 + v^2 = \frac{1}{x^2 + \frac{1}{4}} = -2v$

$u^2 + (v+1)^2 = 1$

即 $|w+i| = 1$

取点 $z_0 = \frac{1}{4}i, w_0 = -4i$

$\therefore$ 映射为 $\text{Im}(w) < 0, |w+i| > 1$



(3) $\operatorname{Re}(z) > 0, 0 < \operatorname{Im}(z) < 1, w = \frac{i}{z}$

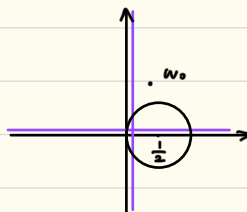
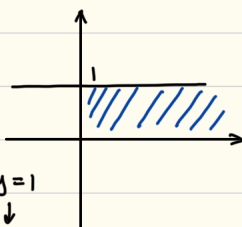
解: 设 $z = x + yi, w = u + iv$

边界 $C_1: x=0 \quad C_2: y=0 \quad C_3: y=1$

$w = \frac{i}{y} (y \neq 0) \quad w = -i\frac{1}{x} \quad w = \frac{1+xi}{x^2+1}, \quad u^2+v^2 = \frac{1}{x^2+1} = u, \quad (u-\frac{1}{2})^2 + v^2 = \frac{1}{4}$
 $\operatorname{Re}(w) = 0 \quad \operatorname{Re}(w) = 0 \quad \operatorname{Re}(w) = 0 \quad |w - \frac{1}{2}| = \frac{1}{2}$

取 $z_0 = 1 + i\frac{1}{2}, w_0 = \frac{2}{3} + \frac{4}{3}i$

$\therefore$ 映射成 $\operatorname{Im}(w) > 0, \operatorname{Re}(w) > 0, |w - \frac{1}{2}| > \frac{1}{2}$



4. 求把上半平面 $\operatorname{Im}(z) > 0$ 映射成单位圆 $|w| < 1$ 的分式线性映射, 且满足 $f(i) = 0, \arg f'(i) = 0$.

解: 令 $z_1 = i, w_1 = 0$.

令 $z_2 = \bar{\lambda}, w_2 = \infty$.

$\therefore w = k \cdot \frac{z - \lambda}{z - \bar{\lambda}} \quad (k \text{ 为复常数}) \quad \dots \text{把上半平面映射成圆的一般形式}$

对于边界点 $|w| = 1, \operatorname{Im}(z) = 0$,

$1 = |k| \cdot \left| \frac{x - \lambda}{x - \bar{\lambda}} \right| \Rightarrow |k| = 1$

$\therefore w = e^{i\theta} \cdot \frac{z - \lambda}{z - \bar{\lambda}} \quad (\operatorname{Im}(\lambda) > 0)$

$\therefore 0 = e^{i\theta} \cdot \frac{i - \lambda}{i - \bar{\lambda}} \quad \text{即 } \lambda = i$

$\therefore w = e^{i\theta} \cdot \frac{z - i}{z + i}$

$w' = e^{i\theta} \cdot \frac{1 \cdot (z+i) - (z-i) \cdot 1}{(z+i)^2} = e^{i\theta} \cdot \frac{2i}{(z+i)^2} \Rightarrow w' = -\frac{1}{2} e^{i(\theta + \frac{\pi}{2})} = \frac{1}{2} e^{i(\theta - \frac{\pi}{2})}, \theta = \frac{\pi}{2}$

$\therefore w = i \frac{z - i}{z + i}$

$\arg w' = 0$

5. 求把单位圆 $|z| < 1$ 映射成单位圆 $|w| < 1$ 的分式线性映射, 且满足条件 $f(\frac{1}{2}) = 0, f(-1) = 1$.

解: 将 $|z| < 1$ 映射成 $|w| < 1$ 的一般形式

$w = e^{i\varphi} \frac{z - \alpha}{1 - \bar{\alpha}z} \quad (|\alpha| < 1, \varphi \in \mathbb{R})$

或 $w = k \frac{z - \alpha}{z - \frac{\alpha}{k}} = k\bar{\alpha} \cdot \frac{z - \alpha}{z - \frac{\alpha}{k}} = (-k\bar{\alpha}) \cdot \frac{z - \alpha}{1 - \bar{\alpha}z}$

$\therefore |z| = 1 \Rightarrow |w| = 1$

$\therefore 1 = |(-k\bar{\alpha})| \cdot \left| \frac{1 - \alpha}{1 - \bar{\alpha}} \right|$

$\therefore |k\bar{\alpha}| = 1,$

$w = e^{i\varphi} \frac{z - \alpha}{1 - \bar{\alpha}z} \quad (|\alpha| < 1)$

$\therefore 0 = e^{i\varphi} \frac{z - \frac{1}{2}}{1 - \bar{\alpha} \cdot \frac{1}{2}}$

$1 = e^{i\varphi} \frac{1 - \frac{1}{2}}{1 - (-1) \cdot \frac{1}{2}} \Rightarrow 1 = e^{i(\varphi - \pi)}, \varphi = \pi$

$\therefore w = - \frac{z - \frac{1}{2}}{1 - \bar{\alpha} \cdot \frac{1}{2}} = - \frac{2z - 1}{2 - z} = \frac{2z - 1}{z - 2}$